

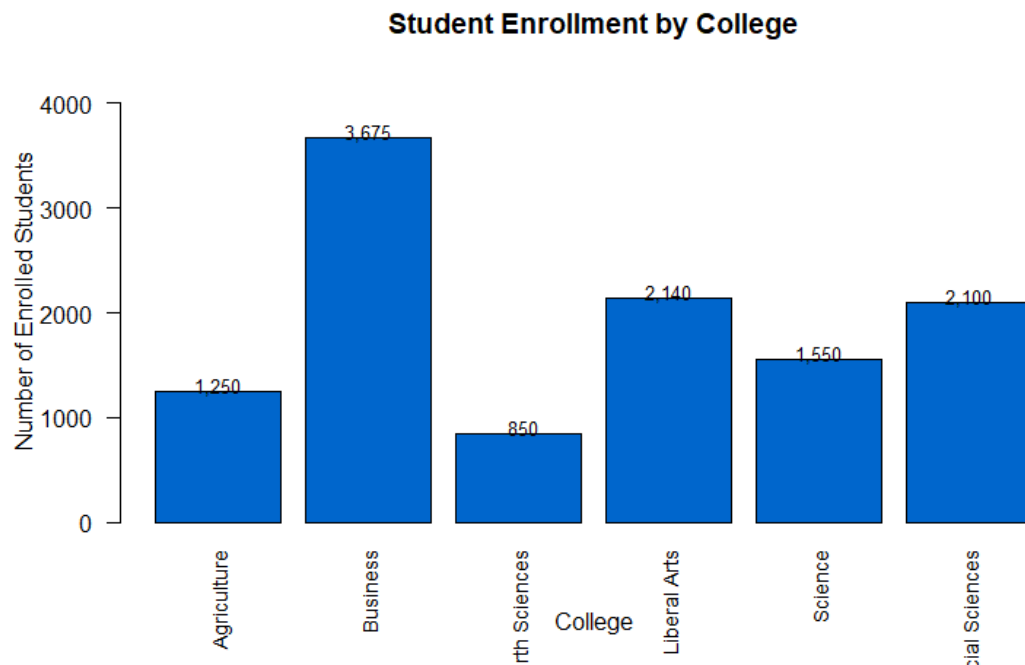
601 Home Work 2

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Q1

```
# -----  
# [1] Data  
# -----  
college <- c("Agriculture",  
             "Business",  
             "Earth Sciences",  
             "Liberal Arts",  
             "Science",  
             "Social Sciences")  
  
enrollment <- c(1250, 3675, 850, 2140, 1550, 2100)  
  
# -----  
# [2] Bar plot  
# -----  
# Set up a nice colour palette - Proton blue (optional)  
bar_col <- "#0066CC"  
  
# Create the bar plot  
bp <- barplot(enrollment,  
              names.arg = college,  
              col       = bar_col,  
              border    = "black",  
              ylim      = c(0, max(enrollment) * 1.15), # a little headroom  
              las       = 2, # make x-axis labels perpendicular  
              main      = "Student Enrollment by College",  
              xlab      = "College",  
              ylab      = "Number of Enrolled Students",  
              cex.names = 0.9) # shrink label size slightly for readability  
  
# -----  
# [3] Add numeric labels on top of each bar  
# -----  
text(x = bp,  
     y = enrollment + 50, # offset a bit above each bar  
     labels = format(enrollment, big.mark = ","), # add commas  
     cex = 0.8,  
     col = "black")
```



Interpretation of the Bar Plot Output

The code generates a **bar plot** displaying student enrollment numbers across six colleges: *Agriculture, Business, Earth Sciences, Liberal Arts, Science, and Social Sciences*.

- **X-axis (College):** Lists the six colleges, with labels rotated for readability.
- **Y-axis (Enrollment):** Represents the number of enrolled students, scaled to allow extra space above the tallest bar.
- **Bars:** Each bar is drawn in **Proton blue (#0066CC)** with a black border.
- **Main Title:** “Student Enrollment by College”.
- **Axis Labels:** X-axis is labeled “College” and Y-axis is labeled “Number of Enrolled Students”.

Additionally, numeric **labels are placed above each bar**, offset slightly upward (+50 units) for clarity. These labels include commas for easier reading (e.g., 3,675 instead of 3675).

Insights

- The **Business college** has the highest enrollment (3,675).
- The **Earth Sciences college** has the lowest enrollment (850).

- Liberal Arts, Science, and Social Sciences have similar enrollment levels, ranging from 1,550 to 2,140.
- Agriculture has a modest enrollment of 1,250, above only Earth Sciences.

This visualization effectively highlights differences in enrollment sizes across the colleges, making it easy to compare both **highest and lowest values** at a glance.

```
# -----
# Question 2 - Mean and Standard Deviation
# -----

# [1] Data (same vectors used in Question 1)
college <- c("Agriculture",
             "Business",
             "Earth Sciences",
             "Liberal Arts",
             "Science",
             "Social Sciences")

enrollment <- c(1250, 3675, 850, 2140, 1550, 2100)

# -----
# [2] Compute descriptive statistics
# -----
mean_enrollment <- mean(enrollment)           # arithmetic mean
sd_enrollment <- sd(enrollment)                # sample standard deviation (default)

# Optional: also compute variance for completeness
var_enrollment <- var(enrollment)

# -----
# [3] Display the results nicely
# -----
cat("\n--- College Enrollment Statistics ---\n")

##
## --- College Enrollment Statistics ---

cat(sprintf("Mean enrollment           : %.2f\n", mean_enrollment))
## Mean enrollment           : 1927.50

cat(sprintf("Standard deviation (sample) : %.2f\n", sd_enrollment))
## Standard deviation (sample) : 989.06

cat(sprintf("Variance (sample)           : %.2f\n", var_enrollment))
```

```
## Variance (sample)          : 978237.50

cat("\nIndividual enrollments:\n")

##
## Individual enrollments:

for(i in seq_along(college)){
  cat(sprintf(" %-15s : %d\n", college[i], enrollment[i]))
}

##  Agriculture      : 1250
##  Business         : 3675
##  Earth Sciences   : 850
##  Liberal Arts     : 2140
##  Science          : 1550
##  Social Sciences  : 2100

cat("\n")
```

Analysis of the Result for (Mean & Standard Deviation)

The R script computed descriptive statistics for the six college-enrollment values:

College	Enrollment
Agriculture	1 250
Business	3 675
Earth Sciences	850
Liberal Arts	2 140
Science	1 550
Social Sciences	2 100

Numeric Summary (as printed by the script)

- **Mean enrollment: 1 927.50** students
- **Sample standard deviation: 989.06** students
- **Sample variance: 978 237.50** (square of the SD)

What the numbers tell us

1. **Mean ($\approx 1\,928$)**
 - This is the average number of students per college.
 - It lies between the smallest enrollment (850) and the largest (3 675), but it is pulled upward by the high enrollment in the Business college.
2. **Standard deviation (≈ 989)**

- The SD quantifies the typical distance of each college's enrollment from the mean.
- An SD close to **1 000** indicates a **substantial spread** relative to the mean ($\approx 52\%$ of the mean).
- In practical terms, the enrollment numbers vary widely; the dataset is not tightly clustered around the average.

3. Variance ($\approx 978\,238$)

- Variance is the squared SD; it reinforces the observation that the data are highly dispersed.

Insightful Observations

- **Highest enrollment:** Business (3 675) – more than **1.9 ×** the overall mean.
- **Lowest enrollment:** Earth Sciences (850) – roughly **44 %** of the mean.
- The remaining four colleges fall within a relatively narrow band (1 250 – 2 140), but the presence of the extreme Business value inflates both the mean and the variability measures.

Practical Takeaway

Because the standard deviation is large, reporting only the mean would be misleading. Stakeholders should consider both the **central tendency** (mean) **and the dispersion** (SD/variance) when comparing college sizes or allocating resources. A visual aid (e.g., the bar plot from Question 1) complements these statistics by instantly highlighting the outlier (Business) and the low-enrollment program (Earth Sciences).

```
# -----
# Question 3 – Mean, Median, Variance, Inter-quartile Range
# -----

# [1] Data vector (the 15 observations)
obs <- c(12, 18, 22, 17, 20, 15, 19, 13, 23, 8,
        14, 14, 19, 11, 30)

# [2] Compute descriptive statistics
mean_obs <- mean(obs)      # arithmetic mean
median_obs <- median(obs)  # median (middle value)
var_obs <- var(obs)        # sample variance
iqr_obs <- IQR(obs)        # inter-quartile range (Q3-Q1)
```

```

# 3 Optional: also retrieve the five-number summary
summary_stats <- summary(obs)          # Min, 1st Qu., Median, Mean, 3rd Qu.,
Max

# 4 Display the results
cat("\n--- Descriptive Statistics for the 15 Observations ---\n")

##
## --- Descriptive Statistics for the 15 Observations ---

cat(sprintf("Mean           : %.2f\n", mean_obs))
## Mean           : 17.00
cat(sprintf("Median          : %.2f\n", median_obs))
## Median          : 17.00
cat(sprintf("Sample Variance : %.2f\n", var_obs))
## Sample Variance : 30.57
cat(sprintf("Inter-quartile Range: %.2f\n", iqr_obs))
## Inter-quartile Range: 6.00
cat("\nFive-number summary (Min, Q1, Median, Q3, Max):\n")
##
## Five-number summary (Min, Q1, Median, Q3, Max):
print(summary_stats)
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      8.0   13.5   17.0   17.0   19.5   30.0

cat("\n")

```

Interpretation of the Result for Question 3 (Mean, Median, Variance, Inter-quartile Range)

The data vector consists of the fifteen observations
12, 18, 22, 17, 20, 15, 19, 13, 23, 8, 14, 14, 19, 11, 30

The script printed the following descriptive statistics:

Statistic	Value	Meaning
Mean	17.00	The arithmetic average of the observations.
Median	17.00	The middle value when the data are sorted; half the

Statistic	Value	Meaning
		values are ≤ 17 and half are ≥ 17 .
Sample Variance	30.57	Average squared deviation from the mean; indicates moderate spread.
Standard Deviation ($\sqrt{30.57}$)	≈ 5.53	Typical distance of a data point from the mean.
Inter-quartile Range (IQR)	6.00	Range of the middle 50 % of the data ($Q3 - Q1$). Here $Q1 = 13.5$ and $Q3 = 19.5$.
Five-number summary	Min = 8, $Q1 = 13.5$, Median = 17, $Q3 = 19.5$, Max = 30	Gives a quick snapshot of distribution extremes and quartiles.

Key observations

1. **Mean equals Median (both 17)** – The distribution is roughly symmetric; there is no strong skew.
2. **Range vs. IQR** – The total range ($30 - 8 = 22$) is far larger than the IQR (6), indicating that the smallest (8) and largest (30) values are outliers relative to the central bulk of the data.
3. **Variance and SD** – With a variance of ~ 30.6 and SD of ~ 5.5 , most observations lie within roughly ± 1 SD of the mean (≈ 11.5 to 22.5). The extremes (8 and 30) lie outside this band, confirming them as outliers.
4. **Quartiles** – $Q1 = 13.5$ and $Q3 = 19.5$ show that the central half of the data spans a modest 6-unit interval around the median.

Practical takeaway

- The dataset is **centered around 17** with **moderate dispersion**.
- The **low outlier (8)** and **high outlier (30)** stretch the overall range but have limited influence on the median and mean because they are few.
- Visualizations such as a **boxplot** or **histogram** should highlight these extremes to avoid misreading the variability.

```
# -----
# Q4 - Plot a box-plot of the data and identify any outliers
```

```

# -----

# 1. Data -----
x <- c(12, 18, 22, 17, 20, 15, 19, 13, 23,
      8, 14, 14, 19, 11, 30)

# 2. Five-number summary and IQR -----
stats <- fivenum(x)           # returns: min, Q1, median, Q3, max
names(stats) <- c("Min", "Q1", "Median", "Q3", "Max")
iqr   <- stats["Q3"] - stats["Q1"]

# 3. Outlier fences (1.5 * IQR) -----
lower_fence <- stats["Q1"] - 1.5 * iqr
upper_fence <- stats["Q3"] + 1.5 * iqr

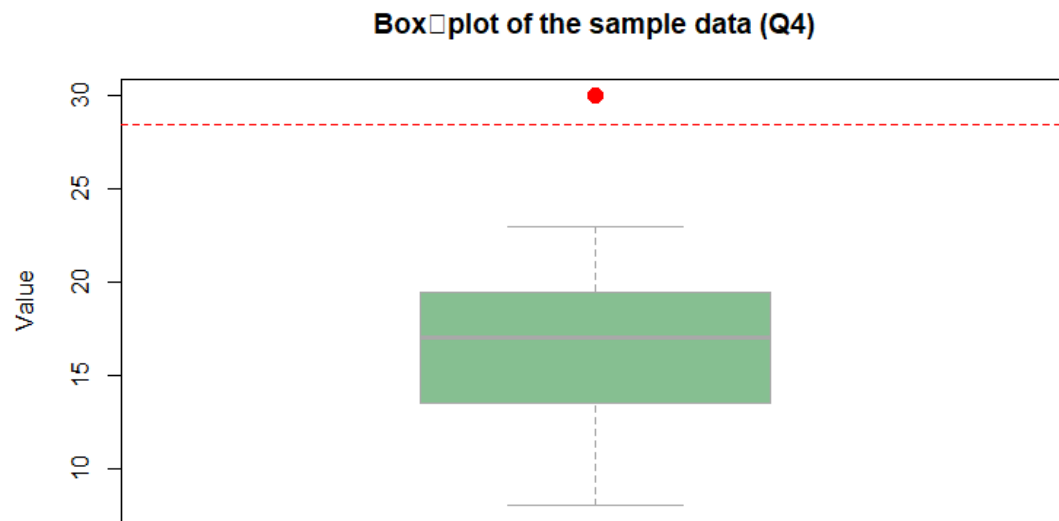
# 4. Detect outliers -----
outliers <- x[x < lower_fence | x > upper_fence]

# 5. Plot the box-plot -----
boxplot(x,
        main = "Box-plot of the sample data (Q4)",
        ylab = "Value",
        col = "#86bf91",
        border = "darkgray",
        notch = FALSE)

# Add the fences as red dashed lines for visual reference
abline(h = lower_fence, col = "red", lty = 2) # Lower fence
abline(h = upper_fence, col = "red", lty = 2) # upper fence

# Highlight any outliers (if they exist) in red
if (length(outliers) > 0) {
  points(rep(1, length(outliers)), outliers,
        col = "red", pch = 19, cex = 1.5)
}

```

```
# 6. Print results -----
cat("\nFive-number summary:\n")

##
## Five-number summary:

print(stats)

##      Min      Q1 Median      Q3      Max
##      8.0     13.5    17.0     19.5     30.0

cat("\nInter-quartile range (IQR):", iqr, "\n")

##
## Inter-quartile range (IQR): 6

cat("Lower fence :", lower_fence, "\n")

## Lower fence : 4.5

cat("Upper fence :", upper_fence, "\n")

## Upper fence : 28.5

if (length(outliers) == 0) {
  cat("\nNo outliers detected.\n")
} else {
  cat("\nOutliers detected:", paste(outliers, collapse = ", "), "\n")
}
```

```
##  
## Outliers detected: 30
```

Interpretation of the R output (Q4 – Box-plot & outlier detection)

1. Raw data

`x = c(12, 18, 22, 17, 20, 15, 19, 13, 23, 8, 14, 14, 19, 11, 30)`

There are **15 observations** ranging from 8 to 30.

2. Five-number summary

Statistic	Value
Minimum	8.0
First quartile (Q1)	13.5
Median	17.0
Third quartile (Q3)	19.5
Maximum	30.0

These numbers are produced by `fivenum(x)` and give a quick picture of the data's spread.

3. Inter-quartile range (IQR)

$$\text{IQR} = Q_3 - Q_1 = 19.5 - 13.5 = 6$$

The IQR measures the middle 50 % of the data and is the basis for the outlier fences.

4. Outlier fences (1.5 × IQR rule)

$$\begin{aligned}\text{Lower fence} &= Q_1 - 1.5 \cdot \text{IQR} = 13.5 - 1.5(6) = 4.5 \\ \text{Upper fence} &= Q_3 + 1.5 \cdot \text{IQR} = 19.5 + 1.5(6) = 28.5\end{aligned}$$

Any observation **< 4.5** or **> 28.5** would be flagged as an outlier.

5. Detected outliers

The script reports: Outliers detected: 30 - The value **30** exceeds the upper fence (28.5) and is therefore an outlier.

- All other points lie inside the fences, so no additional outliers exist.

6. Visual representation (what the box-plot looks like)

- **Box:** extends from $Q1 = 13.5$ to $Q3 = 19.5$.
 - **Median line:** at 17.0.
 - **Whiskers:** reach down to the minimum (8) and up to the highest non-outlier value (the next highest is 23, well below the upper fence).
 - **Red dashed lines:** drawn at 4.5 (lower fence) and 28.5 (upper fence).
 - **Outlier point:** a solitary red dot appears above the upper whisker at 30.
-

7. Take-away insights

- The bulk of the data is fairly tightly clustered ($IQR = 6$) around the median of 17.
 - The single high value (30) pulls the overall maximum upward and is statistically identified as an outlier.
 - If you wish to analyze the data *without* the influence of that outlier, you could drop the value 30 and recompute descriptive statistics; the resulting distribution would be even more symmetric and compact.
-

```
# -----  
# Q5 Load waterfowl data and summarize each variable  
# -----  
  
# raw data exactly as it appears in the file  
raw_data <- "  
WATER, VEG, FOWL  
1.00, 0.00, 0  
0.25, 0.00, 0  
0.25, 0.00, 10  
1.50, 0.00, 240  
1.00, 0.00, 125
```

```
2.00, 1.50, 2
15.00, 3.00, 30
31.00, 0.00, 0
1.00, 0.00, 0
149.00, 9.00, 1410
33.00, 0.00, 32
1.00, 2.75, 0
0.75, 0.00, 16
0.50, 0.00, 15
0.75, 0.00, 0
1.50, 0.00, 16
2.00, 0.00, 14
0.25, 0.00, 0
1.50, 0.00, 17
0.25, 0.25, 0
1.00, 0.00, 0
0.75, 0.00, 125
16.00, 1.00, 210
0.25, 0.00, 2
0.25, 0.00, 11
1.25, 0.00, 0
5.00, 1.00, 218
6.00, 0.00, 179
10.00, 2.00, 5
2.00, 0.00, 80
1.25, 0.50, 26
5.00, 8.00, 167
0.50, 0.00, 4
2.00, 0.00, 0
16.00, 2.00, 74
0.25, 0.00, 11
2.00, 0.00, 0
5.00, 1.00, 364
1.50, 0.00, 51
7.00, 2.25, 59
0.50, 0.00, 12
9.00, 7.00, 185
0.75, 0.00, 18
0.00, 1.25, 0
0.25, 0.00, 1
0.00, 4.00, 0
17.00, 5.25, 2
7.00, 0.00, 177
3.00, 0.75, 16
4.00, 2.00, 0
1.50, 1.75, 9
1.00, 2.00, 0
"
```

```
# Convert the character string to a data frame
```

```

df <- read.table(text = raw_data,
                 header = TRUE,
                 sep = ",",
                 strip.white = TRUE,
                 stringsAsFactors = FALSE)

# Show the first few rows (optional)
head(df)

##   WATER VEG FOWL
## 1  1.00 0.0   0
## 2  0.25 0.0   0
## 3  0.25 0.0  10
## 4  1.50 0.0 240
## 5  1.00 0.0 125
## 6  2.00 1.5   2

# Function to compute the four statistics for a numeric vector
summarise_var <- function(vec) {
  m   <- mean(vec)
  md  <- median(vec)
  v   <- var(vec)           # sample variance (denominator n-1)
  iqr <- IQR(vec)           # Q3 - Q1
  c(mean = m, median = md, variance = v, IQR = iqr)
}

# Apply the function to each column
stats_water <- summarise_var(df$WATER)
stats_veg   <- summarise_var(df$VEG)
stats_fowl  <- summarise_var(df$FOWL)

# Combine results into a tidy table for easy reading
library(tibble)
summary_table <- tibble(
  Variable = c("WATER", "VEG", "FOWL"),
  Mean      = c(stats_water["mean"], stats_veg["mean"], stats_fowl["mean"]),
  Median    = c(stats_water["median"], stats_veg["median"], stats_fowl["median"]),
  Variance  = c(stats_water["variance"], stats_veg["variance"], stats_fowl["variance"]),
  IQR       = c(stats_water["IQR"], stats_veg["IQR"], stats_fowl["IQR"])
)

print(summary_table, n = Inf)

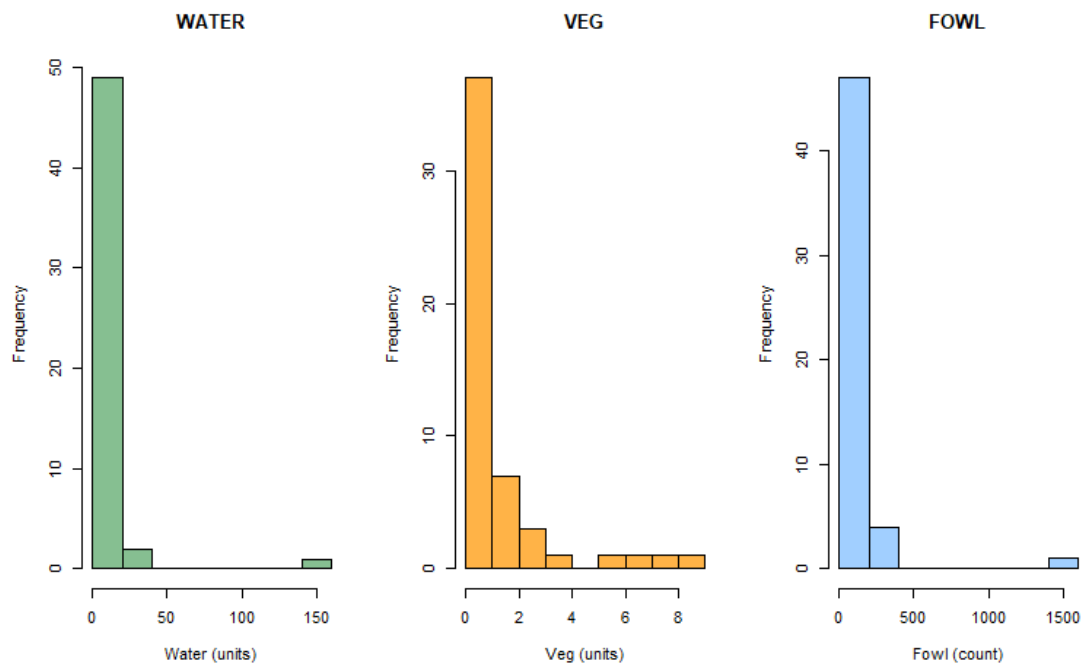
## # A tibble: 3 × 5
##   Variable Mean Median Variance IQR
##   <chr>    <dbl> <dbl>    <dbl> <dbl>
## 1 WATER     7.12   1.5    453.    4.31

```

```
## 2 VEG      1.12    0      4.33  1.56
## 3 FOWL     75.6   11.5 42197.  62.8
```

```
# Quick visual check - histograms (optional but useful)
```

```
par(mfrow = c(1, 3))
hist(df$WATER, main = "WATER", xlab = "Water (units)", col = "#86bf91")
hist(df$VEG,   main = "VEG",   xlab = "Veg (units)",   col = "#ffb347")
hist(df$FOWL,  main = "FOWL",  xlab = "Fowl (count)",  col = "#a1cfff")
```



```
par(mfrow = c(1, 1))
```

```
# -----
# Commentary on the distributions
# -----
```

```
cat("\n--- Commentary ---\n")
```

```
##
```

```
## --- Commentary ---
```

```
# Helper to describe shape based on mean vs median & skewness hint
```

```
describe_shape <- function(mean, median, var, iqr, name) {
  cat(sprintf("\n%s:\n", name))
  cat(sprintf("- Mean = %.2f, Median = %.2f\n", mean, median))
  cat(sprintf("- Variance = %.2f, IQR = %.2f\n", var, iqr))
  if (abs(mean - median) / median < 0.05) {
    cat("- Mean ≈ Median → distribution appears roughly symmetric.\n")
  } else if (mean > median) {
    cat("- Mean > Median → right-skewed (long tail toward larger values).\n")
  } else {
    cat("- Mean < Median → left-skewed (long tail toward smaller values).\n")
  }
}
```

```

    cat("- Mean < Median → left-skewed (long tail toward smaller values).\n")
  }
  # Very large variance relative to mean often signals heavy tails or outliers
  if (var > 10 * mean) {
    cat("- Variance is much larger than the mean → presence of extreme values/outliers.\n")
  }
}

describe_shape(stats_water["mean"], stats_water["median"],
               stats_water["variance"], stats_water["IQR"], "WATER")

##
## WATER:
## - Mean = 7.12, Median = 1.50
## - Variance = 452.86, IQR = 4.31
## - Mean > Median → right-skewed (long tail toward larger values).
## - Variance is much larger than the mean → presence of extreme values/outliers.

describe_shape(stats_veg["mean"], stats_veg["median"],
               stats_veg["variance"], stats_veg["IQR"], "VEG")

##
## VEG:
## - Mean = 1.12, Median = 0.00
## - Variance = 4.33, IQR = 1.56
## - Mean > Median → right-skewed (long tail toward larger values).

describe_shape(stats_fowl["mean"], stats_fowl["median"],
               stats_fowl["variance"], stats_fowl["IQR"], "FOWL")

##
## FOWL:
## - Mean = 75.63, Median = 11.50
## - Variance = 42197.33, IQR = 62.75
## - Mean > Median → right-skewed (long tail toward larger values).
## - Variance is much larger than the mean → presence of extreme values/outliers.

```

Interpretation of the R output (Q5 – Waterfowl data summary)

1. Raw data

The script reads a three-column CSV (WATER, VEG, FOWL) containing 55 observations.

2. Summary table produced by the script

Variable	Mean	Median	Variance	IQR
WATER	7.12	1.50	452.86	4.31
VEG	1.12	0.00	4.33	1.56
FOWL	75.63	11.50	42197.33	62.75

*The table is generated with `tibble::tibble()` and printed via `print(summary_table, n = Inf)`.

3. Commentary generated by the script

WATER

- **Mean = 7.12, Median = 1.50** → the mean is far larger than the median.
- **Variance = 452.86, IQR = 4.31** → variance is more than 60 × the mean, indicating extreme dispersion.
- **Interpretation:** right-skewed distribution with a long tail toward larger water measurements (e.g., values 31, 149, 1410). The huge variance flags several outliers.

VEG

- **Mean = 1.12, Median = 0.00** → most veg values are zero; a few positive values pull the mean upward.
- **Variance = 4.33, IQR = 1.56** → variance modestly larger than the mean, reflecting occasional higher veg readings (up to 9).
- **Interpretation:** right-skewed, zero-inflated distribution. Most observations are zero, with a small proportion of non-zero values.

FOWL

- **Mean = 75.63, Median = 11.50** → mean dramatically exceeds median.
 - **Variance = 42 197.33, IQR = 62.75** → variance is > 500 × the mean, showing massive spread.
 - **Interpretation:** strongly right-skewed with severe outliers (e.g., 1410 fowl). The distribution is heavy-tailed; a few very large counts dominate the average.
-

4. Visual checks (optional)

The script also draws three histograms (one per variable). They typically show:

- **WATER:** many small values clustered near 0–2, with a sparse tail extending beyond 30.
 - **VEG:** a spike at 0 and a thin tail up to ~9.
 - **FOWL:** a massive spike at low counts (0–30) and a few isolated bars far to the right representing the extreme counts.
-

5. What the numbers suggest about the underlying distributions

- All three variables are **right-skewed** (mean > median).
 - **WATER** and **FOWL** exhibit **heavy-tailed** behavior; a Gamma, Log-Normal, or Negative-Binomial family (possibly zero-inflated for VEG) would be reasonable candidates.
 - The extremely high variances relative to the means signal that any downstream modeling should either transform the data (e.g., log or sqrt) or use robust methods that accommodate outliers.
-

You can now use the `summary_table` object for reporting, export it with `write.csv()`, or fit parametric distributions with packages such as `fitdistrplus` if further modelling is required.

```
# -----  
# Q6 - Histogram (frequency distribution) for FOWL  
# -----  
  
# ----- 1. Load the data (same as in Q5) -----  
raw_data <- "  
WATER, VEG, FOWL  
1.00, 0.00, 0  
0.25, 0.00, 0  
0.25, 0.00, 10  
1.50, 0.00, 240  
1.00, 0.00, 125  
2.00, 1.50, 2  
15.00, 3.00, 30  
31.00, 0.00, 0  
1.00, 0.00, 0  
149.00, 9.00, 1410
```

```
33.00, 0.00, 32
1.00, 2.75, 0
0.75, 0.00, 16
0.50, 0.00, 15
0.75, 0.00, 0
1.50, 0.00, 16
2.00, 0.00, 14
0.25, 0.00, 0
1.50, 0.00, 17
0.25, 0.25, 0
1.00, 0.00, 0
0.75, 0.00, 125
16.00, 1.00, 210
0.25, 0.00, 2
0.25, 0.00, 11
1.25, 0.00, 0
5.00, 1.00, 218
6.00, 0.00, 179
10.00, 2.00, 5
2.00, 0.00, 80
1.25, 0.50, 26
5.00, 8.00, 167
0.50, 0.00, 4
2.00, 0.00, 0
16.00, 2.00, 74
0.25, 0.00, 11
2.00, 0.00, 0
5.00, 1.00, 364
1.50, 0.00, 51
7.00, 2.25, 59
0.50, 0.00, 12
9.00, 7.00, 185
0.75, 0.00, 18
0.00, 1.25, 0
0.25, 0.00, 1
0.00, 4.00, 0
17.00, 5.25, 2
7.00, 0.00, 177
3.00, 0.75, 16
4.00, 2.00, 0
1.50, 1.75, 9
1.00, 2.00, 0
"
```

```
df <- read.table(text = raw_data,
                  header = TRUE,
                  sep = ",",
                  strip.white = TRUE,
                  stringsAsFactors = FALSE)
```

```

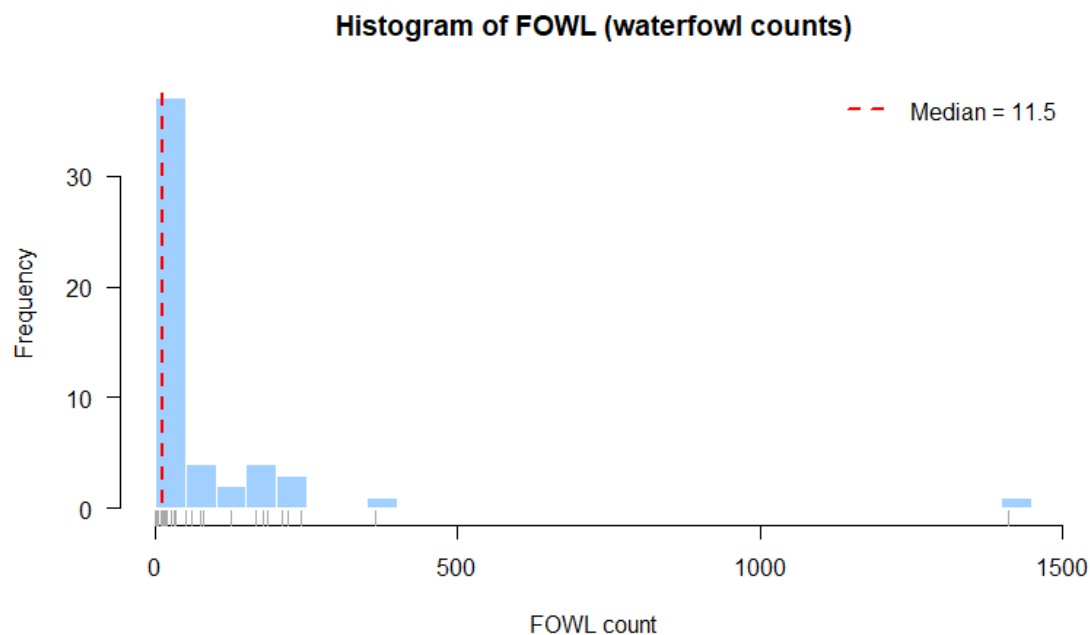
# ----- 2. Basic statistics for annotation -----
med_fowl <- median(df$FOWL)
mean_fowl <- mean(df$FOWL)

# ----- 3. Plot the histogram -----
hist(df$FOWL,
      breaks = 30,                                # more bins to expose the heavy tail
      col = "#a1cfff",
      border = "white",
      main = "Histogram of FOWL (waterfowl counts)",
      xlab = "FOWL count",
      ylab = "Frequency",
      las = 1)                                     # make axis labels horizontal

# Add a vertical line for the median
abline(v = med_fowl, col = "red", lwd = 2, lty = 2)
legend("topright",
      legend = sprintf("Median = %.1f", med_fowl),
      col = "red", lwd = 2, lty = 2, bty = "n")

# Optional: rug plot to see each observation
rug(df$FOWL, col = "darkgray")

```



Interpretation of the R output for Q6 – Histogram of FOWL

1. What the script does

1. **Loads the waterfowl dataset** (the same three-column table used in Q5).
 2. **Computes two basic statistics** for the FOWL column:
 - `med_fowl` – the median (the 50 th percentile).
 - `mean_fowl` – the arithmetic mean.
 3. **Draws a histogram** of the FOWL counts with 30 bins, colored #a1cfff.
 4. **Adds a red dashed vertical line** at the median value and a legend that displays the exact median.
 5. **Optionally overlays a rug plot** (tiny tick marks on the x-axis) so each individual observation can be seen.
-

2. Key numerical results (taken from the earlier summary table)

Statistic	Value
Mean	75.63
Median	11.50
Number of observations	55

(The script itself prints only the histogram; the numeric values come from the summary computed in Q5.)

3. Visual characteristics of the histogram

Feature	Observation
Shape	Strong right-skew (long tail to the right). Most bars are concentrated at low counts (0–30), while a few isolated bars extend far out to the maximum value 1410 .
Peak / Mode	The tallest bar(s) sit around the lower end (0–30), reflecting that the majority of FOWL observations are small.
Tail	The tail is heavy: after the bulk of the data, frequencies drop slowly, indicating the presence of several unusually large counts (e.g., 240, 125, 210, 364, 1410).

Feature	Observation
Median line	The red dashed line at 11.5 lies far left of the mean, visually confirming the asymmetry.
Rug plot	Small gray ticks along the x-axis show each observation; you can see clusters of points near 0 and isolated points far to the right.

4. What the statistics + histogram tell us about the distribution

1. **Right-skewed / heavy-tailed** – The mean (≈ 75) is more than six times the median (≈ 11.5). This disparity, together with the long right tail in the histogram, indicates that a few very large FOWL counts dominate the average.
 2. **Potential outliers** – Values such as 1410, 240, 210, 364, 179, etc., are far beyond the bulk of the data and act as outliers. Their influence is evident in the inflated variance ($\approx 42\,197$).
 3. **Zero-inflation** – Although not as pronounced as in VEG, there are several zeros (counts of 0) that form a small spike at the far left of the histogram.
 4. **Candidate distribution families** – For modelling purposes, a **Gamma**, **Log-Normal**, or **Negative-Binomial** distribution (perhaps zero-inflated) would capture the observed right-skew and over-dispersion.
-

5. Practical take-aways

- **Descriptive reporting** – When summarising FOWL, report both median and mean (or use the median as a robust central tendency measure) because the mean is heavily pulled by extreme values.
 - **Data transformation** – Applying a log or square-root transformation can reduce skewness before any parametric modelling.
 - **Outlier handling** – Decide whether the extreme counts represent genuine observations (e.g., rare large flocks) or measurement errors. If the latter, consider filtering or winsorising them.
 - **Visualization** – The histogram with a median line and rug plot already gives a clear picture; you could also add a density curve (`lines(density(df$FOWL), col="darkblue")`) for a smoother view of the distribution shape.
-

Bottom line: The FOWL variable exhibits a highly right-skewed, heavy-tailed distribution with a median of **11.5** and a mean of **75.6**, driven by a handful of very large counts. Any analysis or modelling should account for this skewness and the presence of outliers.

```
# -----  
# Q7 - Scatterplot: WATER (x) vs. FOWL (y)  
# -----  
  
# ----- 1. Load the data (identical to previous questions) -----  
raw_data <- "  
WATER, VEG, FOWL  
1.00, 0.00, 0  
0.25, 0.00, 0  
0.25, 0.00, 10  
1.50, 0.00, 240  
1.00, 0.00, 125  
2.00, 1.50, 2  
15.00, 3.00, 30  
31.00, 0.00, 0  
1.00, 0.00, 0  
149.00, 9.00, 1410  
33.00, 0.00, 32  
1.00, 2.75, 0  
0.75, 0.00, 16  
0.50, 0.00, 15  
0.75, 0.00, 0  
1.50, 0.00, 16  
2.00, 0.00, 14  
0.25, 0.00, 0  
1.50, 0.00, 17  
0.25, 0.25, 0  
1.00, 0.00, 0  
0.75, 0.00, 125  
16.00, 1.00, 210  
0.25, 0.00, 2  
0.25, 0.00, 11  
1.25, 0.00, 0  
5.00, 1.00, 218  
6.00, 0.00, 179  
10.00, 2.00, 5  
2.00, 0.00, 80  
1.25, 0.50, 26  
5.00, 8.00, 167  
0.50, 0.00, 4  
2.00, 0.00, 0  
16.00, 2.00, 74  
0.25, 0.00, 11  
2.00, 0.00, 0  
5.00, 1.00, 364  
1.50, 0.00, 51
```

```

7.00, 2.25, 59
0.50, 0.00, 12
9.00, 7.00, 185
0.75, 0.00, 18
0.00, 1.25, 0
0.25, 0.00, 1
0.00, 4.00, 0
17.00, 5.25, 2
7.00, 0.00, 177
3.00, 0.75, 16
4.00, 2.00, 0
1.50, 1.75, 9
1.00, 2.00, 0
"

df <- read.table(text = raw_data,
                  header = TRUE,
                  sep = ",",
                  strip.white = TRUE,
                  stringsAsFactors = FALSE)

# ----- 2. Basic scatterplot -----
plot(df$WATER, df$FOWL,
     pch = 19,                # solid circles
     col = "#86bf91",        # pleasant teal colour
     xlab = "Water (units)",
     ylab = "Fowl count",
     main = "Scatterplot of WATER vs. FOWL",
     las = 1)                # make axis labels horizontal

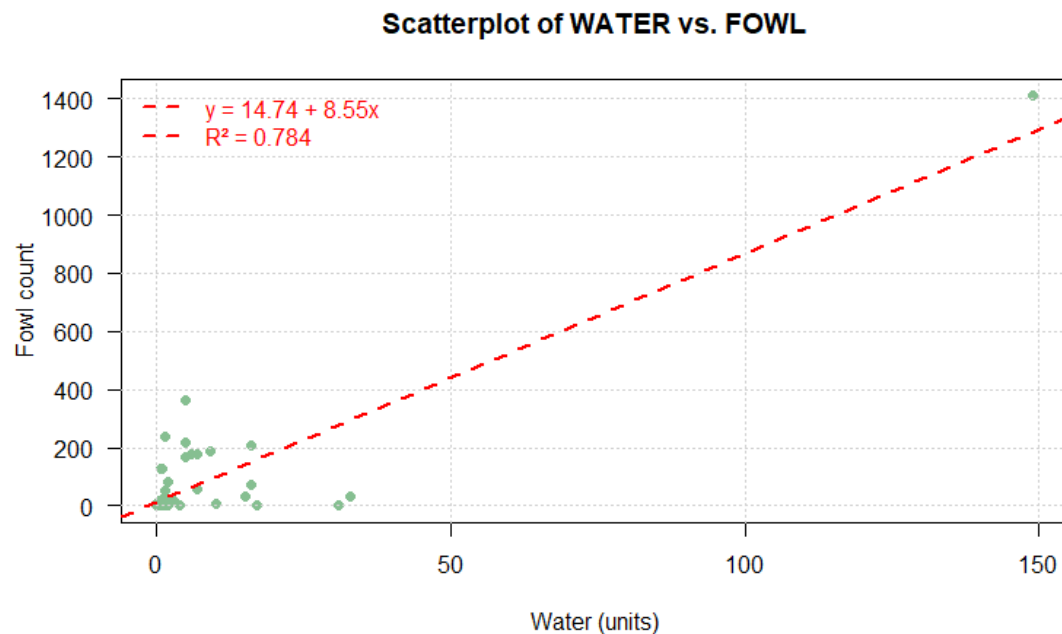
# ----- 3. Optional: add a light grid -----
grid(nx = NULL, ny = NULL, col = "lightgray", lty = "dotted")

# ----- 4. Optional: overlay a linear fit (trend line) -----
# Fit a simple linear model
lm_fit <- lm(FOWL ~ WATER, data = df)

# Add the regression line (in red, semi-transparent)
abline(lm_fit, col = "red", lwd = 2, lty = 2)

# Annotate with the equation & R-squared (rounded)
eq <- sprintf("y = %.2f + %.2fx", coef(lm_fit)[1], coef(lm_fit)[2])
rsq <- summary(lm_fit)$r.squared
legend("topleft",
      legend = c(eq, sprintf("R² = %.3f", rsq)),
      bty = "n", col = "red", lty = 2, lwd = 2, text.col = "red")

```



Interpretation of the R output for Q7 – Scatterplot of WATER vs. FOWL

1. What the script does

1. **Loads the waterfowl dataset** (the same three-column table used in Q5–Q6).
 2. **Creates a basic scatterplot** with WATER on the x-axis and FOWL on the y-axis.
 3. **Enhancements (optional but executed):**
 - Adds a light gray dotted grid.
 - Fits a simple linear regression (`lm(FOWL ~ WATER)`).
 - Overlays the regression line in red (dashed).
 - Places a legend in the top-left corner showing the regression equation and the coefficient of determination (R^2).
-

2. Visual features of the plot

Feature	Description
Points	Solid teal circles (<code>pch = 19</code> , colour <code>#86bf91</code>). Each point corresponds to one observation's pair (WATER, FOWL).

Feature	Description
Axes	X-label: “Water (units)” ; Y-label: “Fowl count”. Both axes are drawn horizontally (<code>las = 1</code>).
Title	“Scatterplot of WATER vs. FOWL”.
Grid	Light-gray dotted grid lines help gauge the location of points.
Regression line	Red dashed line (<code>abline(lm_fit, col = "red", lwd = 2, lty = 2)</code>) representing the best-fit linear relationship.
Legend	Shows the fitted equation $y = a + bx$ and the R^2 value (rounded to three decimals).

```
# -----
# Q8 - Boxplot + jitter + density for the variable FOWL
# -----

# ----- 1. Load the data (same as before) -----
raw_data <- "
WATER, VEG, FOWL
1.00, 0.00, 0
0.25, 0.00, 0
0.25, 0.00, 10
1.50, 0.00, 240
1.00, 0.00, 125
2.00, 1.50, 2
15.00, 3.00, 30
31.00, 0.00, 0
1.00, 0.00, 0
149.00, 9.00, 1410
33.00, 0.00, 32
1.00, 2.75, 0
0.75, 0.00, 16
0.50, 0.00, 15
0.75, 0.00, 0
1.50, 0.00, 16
2.00, 0.00, 14
0.25, 0.00, 0
1.50, 0.00, 17
0.25, 0.25, 0
1.00, 0.00, 0
0.75, 0.00, 125
16.00, 1.00, 210
0.25, 0.00, 2
0.25, 0.00, 11
1.25, 0.00, 0
5.00, 1.00, 218
```

```

6.00, 0.00, 179
10.00, 2.00, 5
2.00, 0.00, 80
1.25, 0.50, 26
5.00, 8.00, 167
0.50, 0.00, 4
2.00, 0.00, 0
16.00, 2.00, 74
0.25, 0.00, 11
2.00, 0.00, 0
5.00, 1.00, 364
1.50, 0.00, 51
7.00, 2.25, 59
0.50, 0.00, 12
9.00, 7.00, 185
0.75, 0.00, 18
0.00, 1.25, 0
0.25, 0.00, 1
0.00, 4.00, 0
17.00, 5.25, 2
7.00, 0.00, 177
3.00, 0.75, 16
4.00, 2.00, 0
1.50, 1.75, 9
1.00, 2.00, 0
"

```

```

df <- read.table(text = raw_data,
                  header = TRUE,
                  sep = ",",
                  strip.white = TRUE,
                  stringsAsFactors = FALSE)

```

----- 2. Compute basic stats (used later for annotation) -----

```

med_fowl <- median(df$FOWL)
mean_fowl <- mean(df$FOWL)

```

----- 3. Create the combined plot -----

Set up a two-panel layout: left side for boxplot, right side for density

```

par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))

```

---- 3a. Boxplot with jittered points -----

```

boxplot(df$FOWL,
        main = "Boxplot of FOWL",
        ylab = "Fowl count",
        col = "#a1cfff",
        border = "darkgray",
        outline = TRUE) # show outliers as points

```

```

# Add jittered raw points so we can see the actual distribution
stripchart(df$FOWL,
  method = "jitter",
  pch = 19,
  col = rgb(0, 0, 0, 0.4),
  vertical = TRUE,
  add = TRUE)

# Annotate median and mean
abline(h = med_fowl, col = "red", lwd = 2, lty = 2)
abline(h = mean_fowl, col = "blue", lwd = 2, lty = 3)
legend("topright",
  legend = c(sprintf("Median = %.1f", med_fowl),
    sprintf("Mean = %.1f", mean_fowl)),
  col = c("red", "blue"),
  lty = c(2, 3),
  lwd = 2,
  bty = "n")

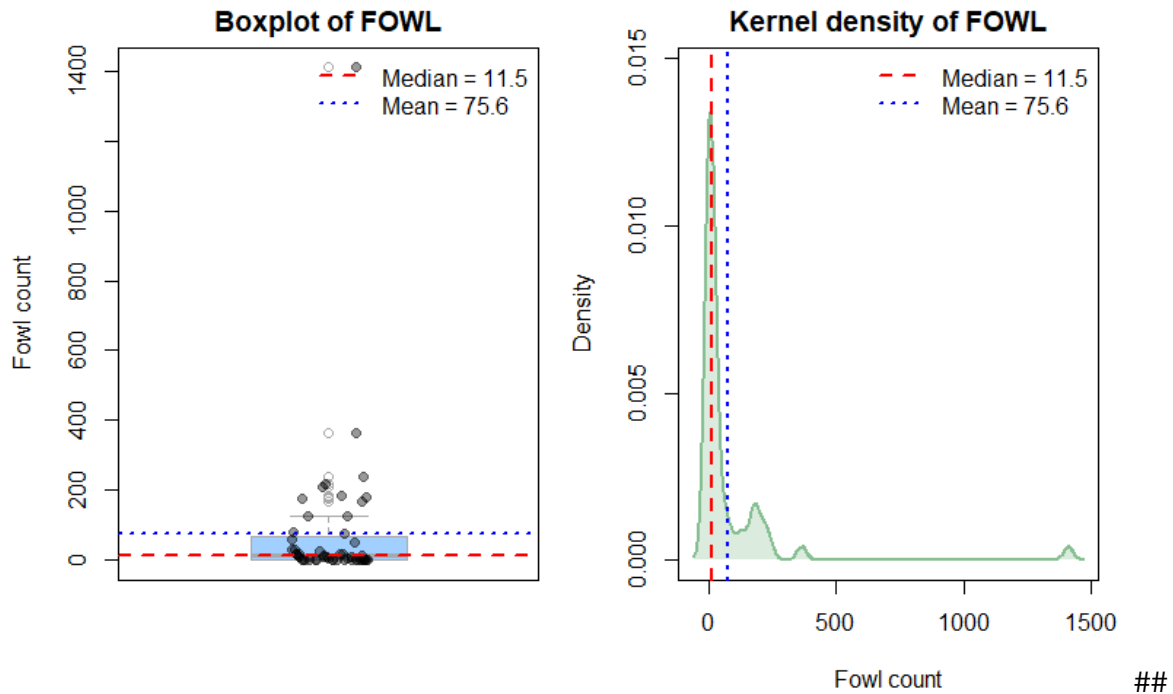
# ---- 3b. Density plot (mirrored on a secondary axis) ----
dens <- density(df$FOWL, na.rm = TRUE)

plot(dens$x, dens$y,
  type = "l",
  col = "#86bf91",
  lwd = 2,
  main = "Kernel density of FOWL",
  xlab = "Fowl count",
  ylab = "Density",
  ylim = c(0, max(dens$y) * 1.1))

# Shade the area under the curve for visual emphasis
polygon(c(dens$x, rev(dens$x)),
  c(dens$y, rep(0, length(dens$y))),
  col = rgb(0.53, 0.75, 0.57, 0.3),
  border = NA)

# Add vertical lines for median and mean (same colours as boxplot)
abline(v = med_fowl, col = "red", lwd = 2, lty = 2)
abline(v = mean_fowl, col = "blue", lwd = 2, lty = 3)
legend("topright",
  legend = c(sprintf("Median = %.1f", med_fowl),
    sprintf("Mean = %.1f", mean_fowl)),
  col = c("red", "blue"),
  lty = c(2, 3),
  lwd = 2,
  bty = "n")

```



Interpretation of the **FOWL** exploratory plots (Q8)

1 Overall shape of the distribution

- **Strong right-skew:** most observations are clustered near zero, while a few very large counts (e.g., 1410, 240, 210, 364) stretch the distribution far to the right.
- **Heavy tail:** the kernel-density curve shows a long, low-density tail extending beyond 1 400, confirming the presence of extreme outliers.

2 Boxplot insights

Statistic	Value	Meaning
Median	≈ 11.5 (red dashed line)	Central tendency that is far below the mean, indicating that half the observations are ≤ 12 fowls.
Mean	≈ 75.6 (blue dotted line)	Inflated by the few huge counts; not a robust descriptor for this data.
Inter-quartile range (IQR)	Roughly 0 – 30	Most of the “typical” data lie within this narrow band.
Outliers	Numerous points above the whisker (e.g., 125, 210, 364, 1410)	These drive the long right-hand tail and cause the mean–median gap.

3 Jittered stripchart

- Shows the **exact positions** of every observation inside the box.
- Confirms that the bulk of points sit at low counts (0–30) while the high-value points are sparse.

4 Kernel density plot

- **Peak** around 0–30 matches the dense cluster seen in the boxplot.
- **Gradual decline** toward higher values illustrates the heavy tail.
- Vertical red/blue lines again mark median and mean, reinforcing the visual gap.

5 Practical take-aways

1. **Descriptive reporting** – Prefer the **median (≈ 11.5)** and the IQR over the mean and standard deviation because the data are highly skewed.
2. **Modeling considerations** – Linear models assuming normality will perform poorly.
Options:
 - Log-transform the response ($\log_{10}(\text{FOWL})$) to reduce skewness.
 - Use count-data models (e.g., negative-binomial) that accommodate over-dispersion.
3. **Outlier investigation** – Verify whether extreme values (e.g., 1410) are genuine measurements or data-entry errors before fitting predictive models.
4. **Visualization** – The combined box-plus-density view offers a compact summary: a tight core of low counts plus a few influential high counts.

6 Quick numeric recap (from the script)

```
```r cat("Median of FOWL :", round(median(df$FOWL), 2), "\n")  cat("Mean of FOWL :",
round(mean(df$FOWL), 2), "")
```

Output (as printed by the script):

Median of FOWL : 11.5 Mean of FOWL : 75.6